

Program: Diploma in Renewable Energy	
Course Code: 3299	Course Title: Basic Renewable Energy Workshop
Semester: 3	Credits: 1.5
Course Category: Program Core Course	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To familiarize with the design of electrical installation systems in residential buildings.
- To provide a basic idea of solar PV interconnection.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic Concept of electrical systems		Fundamentals of Electrical & Electronics Engineering	2
Basic concepts of renewable Energy systems		Basic Electrical Engineering & Renewable Technology	2

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify various wiring systems and the usage of appropriate accessories for electrification.	6	Applying
CO2	Develop and practice different electrical wiring installations as per IS standard	12	Applying

CO3	Identify the basic tests conducted for electrical installations	9	Applying
CO4	Develop and practice various interconnections of solar PV systems.	12	Applying
	Lab Test	6	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2				3			3
CO3				3			3
CO4				3	2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Identify various wiring systems and the usage of appropriate accessories for electrification.		
M1.01	Practice various safety precautions and first aid.	1	Applying
M1.02	Identify different types of wiring, wiring accessories and draw electrical symbols	2	Understanding
M1.03	Practice the usage of wire gauge, line tester, Clamp-on ammeter, test lamp, extension board and series testing board.	2	Applying
M1.04	Select suitable electric wires for solar applications such as wiring, measuring and testing.	1	Applying

CO2	Develop and practice different electrical wiring installations as per IS standard		
M2.01	Model a system to control two lamps controlled by two switches independently (Use PVC surface conduit wiring system)	3	Applying
M2.03	Model a standard stair case wiring system (Use PVC surface conduit wiring system)	3	Applying
M2.04	Develop the wiring installation for hospital wiring (Use PVC surface conduit wiring system)	3	Applying
M2.05	Make use of PVC casing and capping wiring systems to build go-down wiring.	3	Applying
	Lab Test – I	3	
CO3	Identify the basic tests conducted for electrical installations		
M3.01	Develop a meter board for lighting installation using energy meter, fuse, MCB, DP switch, ELCB and indicator.	3	Applying
M3.02	Utilize megger to test the insulation of wiring.	3	Applying
M3.03	Demonstrate pipe/plate earthing and measure the earth resistance	3	Applying
C04	Develop and practice various interconnections of solar PV systems.		
M4.01	Familiarize the parts, roofing techniques and installation of solar panels.	3	Understanding
M4.02	Construct a PV array using series connected solar panels and check the output.	3	Applying
M4.03	Construct a PV array using parallel connected solar panels and check the output.	3	Applying
M4.04	Construct a PV array using a series -parallel combination of solar panels and check the output.	3	Applying

	Lab Test – II	3	
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Text / Reference

T/R	Book Title/Author
T1	The Complete Lab Manual For Renewable Energy - Cutter , Ammond - Delmar Cengage Learning
T2	Electrical wiring and estimating and costing -S L Uppel -Khanna publications
R1	Electrical Design Estimating and Costing, K. B. Raina, New Age International
R2	Fundamentals of maintenance of electrical equipment -K .B Bhatia Khanna Publications
R3	Electric Wiring for Domestic Installers- Brian Scaddan- Publisher: Routledge

Online Resources

Sl.No	Website Link
1	Hands on experiments : https://files.eric.ed.gov/fulltext/EJ1057057.pdf
2	https://vlab.amrita.edu/index.php?sub=77&brch=298
3	https://renewablelab.niu.edu/experiments/seriesParallelSolarCells
4	https://web.iitd.ac.in/~pmvs/courses/mcl241/laboratory-manual-RES.pdf
5	https://www.coursera.org/lecture/photovoltaic-systems/module-components-skzP1
6	https://electrical-engineering-portal.com/photovoltaic-module-interconnections

Student Activity

Suggested Open-ended Experiments:

Students can do open ended experiments as a group of 3-5. There is no duplication in experiments between groups. This is mainly for the purpose of continuous internal evaluation

and a score of 15 marks. Students should prepare a separate report on the open ended experiment of their choice.

Example: 1.

Model a simple product (Mobile Charger) to demonstrate how solar systems can help to improve life?

Develop a mini solar panel doing wiring and soldering of small solar cells

Identify various faults occurring in an iron box (ordinary) and prepare the maintenance chart of the possible faults and their remedies